Pregroup representable expansions of residuated lattices

Andrew Craig Claudette Robinson

RAMiCS 2026
7–10 April
Będlewo, Poland

Relation algebras

Relation algebras

$\mathbf{A} = \langle A, \wedge, \vee, ', \bot, \top, \cdot, 1, ^{\sqcup} \rangle$ is a relation algebra if

- $\langle A, \wedge, \vee, ', \bot, \top \rangle$ is a Boolean algebra
- $\langle A, \cdot, 1 \rangle$ is a monoid

Relation algebras

$\mathbf{A} = \langle A, \wedge, \vee, ', \bot, \top, \cdot, 1, ^{\sqcup} \rangle$ is a relation algebra if

- $\langle A, \wedge, \vee, ', \bot, \top \rangle$ is a Boolean algebra
- $\langle A, \cdot, 1 \rangle$ is a monoid

and

1. $x^{\sqcup\sqcup} = x$
2. $(xy)^{\sqcup} = y^{\sqcup} x^{\sqcup}$
3. $x(y \vee z) = xy \vee xz$
4. $(x \vee y)^{\sqcup} = x^{\sqcup} \vee y^{\sqcup}$
5. $x^{\sqcup}(xy)' \leqslant y'$

Relation algebras

$\mathbf{A} = \langle A, \wedge, \vee, ', \bot, \top, \cdot, 1, ^{\sqcup} \rangle$ is a relation algebra if

- $\langle A, \wedge, \vee, ', \bot, \top \rangle$ is a Boolean algebra
- $\langle A, \cdot, 1 \rangle$ is a monoid

and

1. $x^{\sqcup\sqcup} = x$
2. $(xy)^{\sqcup} = y^{\sqcup}x^{\sqcup}$
3. $x(y \vee z) = xy \vee xz$
4. $(x \vee y)^{\sqcup} = x^{\sqcup} \vee y^{\sqcup}$
5. $x^{\sqcup}(xy)' \leqslant y'$

Let X be a set and $\mathrm{id}_X = \{(x, x) \mid x \in X\}$. Then

$$\mathcal{A}(X) = \langle \wp(X^2), \cap, \cup, ^{\mathsf{c}}, \emptyset, X^2, ;, \mathrm{id}_X, ^{\smile} \rangle$$

is a relation algebra.

Relation algebras

$\mathbf{A} = \langle A, \wedge, \vee, ', \bot, \top, \cdot, 1, ^{\sqcup} \rangle$ is a relation algebra if

- $\langle A, \wedge, \vee, ', \bot, \top \rangle$ is a Boolean algebra
- $\langle A, \cdot, 1 \rangle$ is a monoid

and

1. $x^{\sqcup\sqcup} = x$
2. $(xy)^{\sqcup} = y^{\sqcup} x^{\sqcup}$
3. $x(y \vee z) = xy \vee xz$
4. $(x \vee y)^{\sqcup} = x^{\sqcup} \vee y^{\sqcup}$
5. $x^{\sqcup}(xy)' \leqslant y'$

Let X be a set and $\mathrm{id}_X = \{(x, x) \mid x \in X\}$. Then

$$\mathcal{A}(X) = \langle \wp(X^2), \cap, \cup, ^{\mathsf{c}}, \emptyset, X^2, ;, \mathrm{id}_X, ^{\smile} \rangle$$

is a relation algebra.
Relation algebra $\mathbf{A}$ is representable if $\mathbf{A} \in \mathbb{ISP}(\{\mathcal{A}(X) \mid X \text{ a set}\})$

Group representable RAs

Let $\mathbf{G} = \langle G, \cdot, ^{-1}, e \rangle$ be a group.

Group representable RAs

Let $\mathbf{G} = \langle G, \cdot, ^{-1}, e \rangle$ be a group. Consider $\wp(G)$ and $A, B \subseteq G$:

- $A \bullet B = \{ a \cdot b \mid a \in A, b \in B \}$
- $A^{-1} = \{ a^{-1} \mid a \in A \}$

Group representable RAs

Let $\mathbf{G} = \langle G, \cdot, ^{-1}, e \rangle$ be a group. Consider $\wp(G)$ and $A, B \subseteq G$:

- $A \bullet B = \{ a \cdot b \mid a \in A, b \in B \}$
- $A^{-1} = \{ a^{-1} \mid a \in A \}$

Theorem (McKinsey 1940s)

If $\mathbf{G}$ *is a group then* $\mathcal{R}(\mathbf{G}) = \langle \wp(G), \cap, \cup, ^{\mathsf{c}}, \emptyset, G, \bullet, \{e\}, ^{-1} \rangle$ *is a relation algebra.*

Group representable RAs

Let $\mathbf{G} = \langle G, \cdot, ^{-1}, e \rangle$ be a group. Consider $\wp(G)$ and $A, B \subseteq G$:

- $A \bullet B = \{ a \cdot b \mid a \in A, b \in B \}$
- $A^{-1} = \{ a^{-1} \mid a \in A \}$

Theorem (McKinsey 1940s)

If $\mathbf{G}$ *is a group then* $\mathcal{R}(\mathbf{G}) = \langle \wp(G), \cap, \cup, ^{\mathsf{c}}, \emptyset, G, \bullet, \{e\}, ^{-1} \rangle$ *is a relation algebra.*

Theorem (Jónsson & Tarski 1952)

If $\mathbf{A} \in$ ***RA*** *and* $\mathbf{A} \approx \mathcal{R}(\mathbf{G})$ *for some group* $\mathbf{G}$*, then* $\mathbf{A} \in$ ***RRA****.*

Group representable RAs

Let $\mathbf{G} = \langle G, \cdot, ^{-1}, e \rangle$ be a group. Consider $\wp(G)$ and $A, B \subseteq G$:

- $A \bullet B = \{ a \cdot b \mid a \in A, b \in B \}$
- $A^{-1} = \{ a^{-1} \mid a \in A \}$

Theorem (McKinsey 1940s)

If $\mathbf{G}$ *is a group then* $\mathcal{R}(\mathbf{G}) = \langle \wp(G), \cap, \cup, ^{\mathsf{c}}, \emptyset, G, \bullet, \{e\}, ^{-1} \rangle$ *is a relation algebra.*

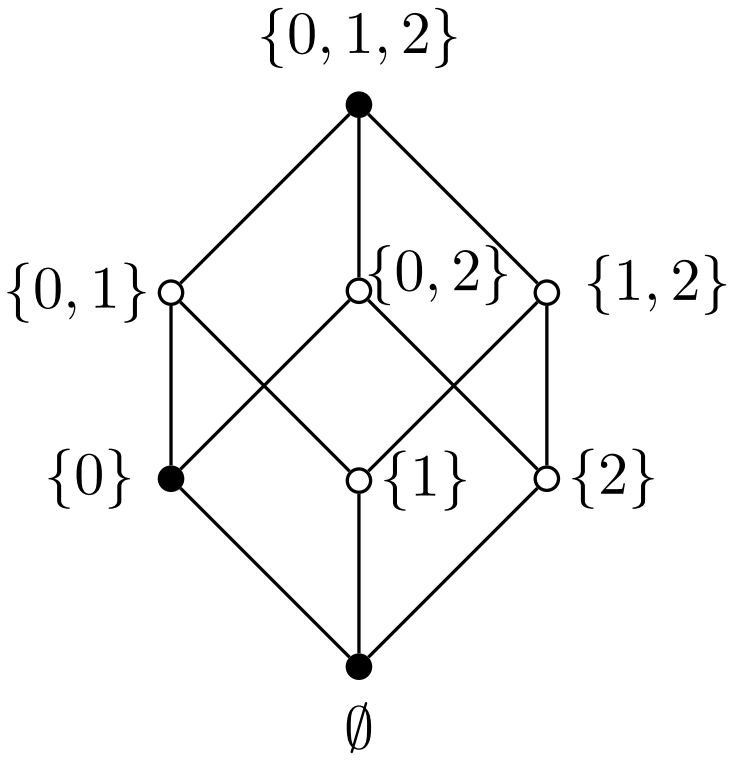
Theorem (Jónsson & Tarski 1952)

If $\mathbf{A} \in$ ***RA*** *and* $\mathbf{A} \approx \mathcal{R}(\mathbf{G})$ *for some group* $\mathbf{G}$*, then* $\mathbf{A} \in$ ***RRA***.

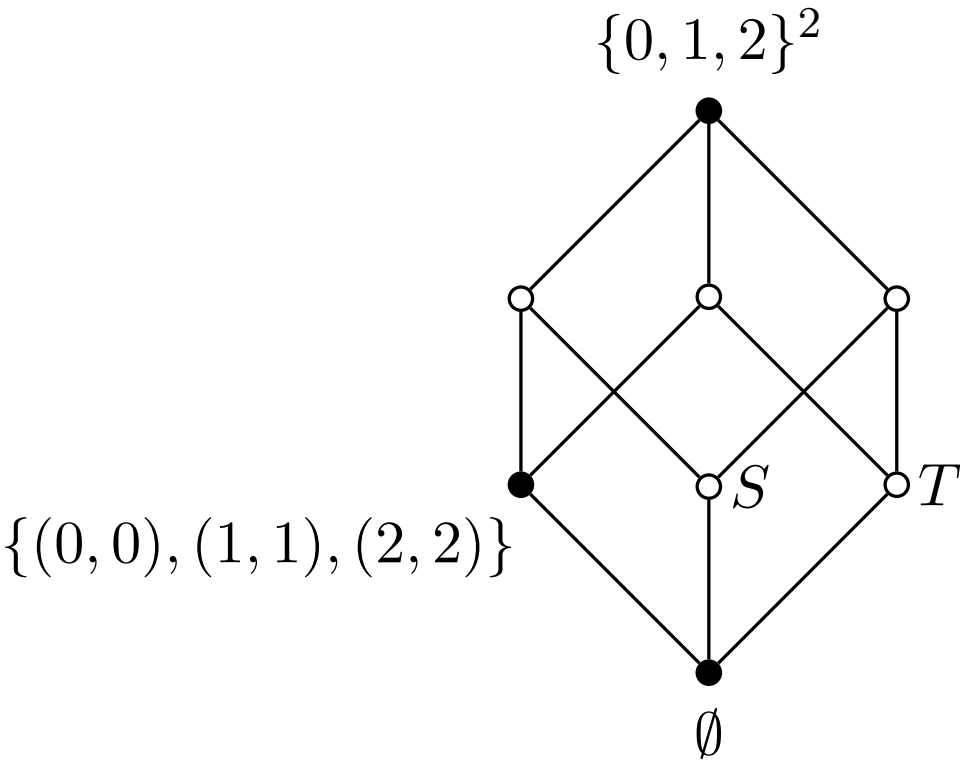
Proof.

The map $\sigma : \wp(G) \to \wp(G \times G)$ defined by $\sigma(Y) = \{ (g, g \cdot y) \mid g \in G, y \in Y \}$ is an embedding of $\mathcal{R}(\mathbf{G})$ into $\mathcal{A}(G)$. $\square$

Example: group representable relation algebra



$\mathcal{R}(\mathbb{Z}_3)$



$$S = \{(0,1),(1,2),(2,0)\}$$
$$T = \{(0,2),(1,0),(2,1)\}$$

$\sigma(\mathcal{R}(\mathbb{Z}_3))$

Generalisations of relation algebras

$\mathbf{A} = \langle A, \wedge, \vee, \cdot, 1, \backslash, / \rangle$ is a residuated lattice (RL) if $\langle A, \wedge, \vee \rangle$ is a lattice and $\langle A, \cdot, 1 \rangle$ is a monoid such that:

$$a \cdot b \leqslant c \quad \Longleftrightarrow \quad b \leqslant a \backslash c \quad \Longleftrightarrow \quad a \leqslant c / b.$$

Generalisations of relation algebras

$\mathbf{A} = \langle A, \wedge, \vee, \cdot, 1, \backslash, / \rangle$ is a residuated lattice (RL) if $\langle A, \wedge, \vee \rangle$ is a lattice and $\langle A, \cdot, 1 \rangle$ is a monoid such that:

$$a \cdot b \leqslant c \quad \Longleftrightarrow \quad b \leqslant a \backslash c \quad \Longleftrightarrow \quad a \leqslant c / b.$$

Definition (cf. Galatos et al. 2007)

Let $\mathbf{A}$ be a residuated lattice and $0 \in A$. Then $\langle A, \wedge, \vee, \cdot, 1, 0, \backslash, / \rangle$ is an FL-algebra.

Generalisations of relation algebras

$\mathbf{A} = \langle A, \wedge, \vee, \cdot, 1, \backslash, / \rangle$ is a residuated lattice (RL) if $\langle A, \wedge, \vee \rangle$ is a lattice and $\langle A, \cdot, 1 \rangle$ is a monoid such that:

$$a \cdot b \leqslant c \quad \Longleftrightarrow \quad b \leqslant a \backslash c \quad \Longleftrightarrow \quad a \leqslant c / b.$$

Definition (cf. Galatos et al. 2007)

Let $\mathbf{A}$ be a residuated lattice and $0 \in A$. Then $\langle A, \wedge, \vee, \cdot, 1, 0, \backslash, / \rangle$ is an FL-algebra. Define

$$\sim a = a \backslash 0 \qquad \text{and} \quad -a = 0/a.$$

If $\sim - a = a = - \sim a$ for all $a \in A$, then $\langle A, \wedge, \vee, \cdot, 1, 0, \backslash, / \rangle$ is an InFL-algebra.

Generalisations of relation algebras

$\mathbf{A} = \langle A, \wedge, \vee, \cdot, 1, \backslash, / \rangle$ is a residuated lattice (RL) if $\langle A, \wedge, \vee \rangle$ is a lattice and $\langle A, \cdot, 1 \rangle$ is a monoid such that:

$$a \cdot b \leqslant c \quad \Longleftrightarrow \quad b \leqslant a \backslash c \quad \Longleftrightarrow \quad a \leqslant c / b.$$

Definition (cf. Galatos et al. 2007)

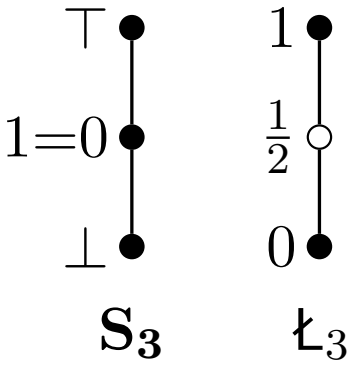
Let $\mathbf{A}$ be a residuated lattice and $0 \in A$. Then $\langle A, \wedge, \vee, \cdot, 1, 0, \backslash, / \rangle$ is an FL-algebra. Define

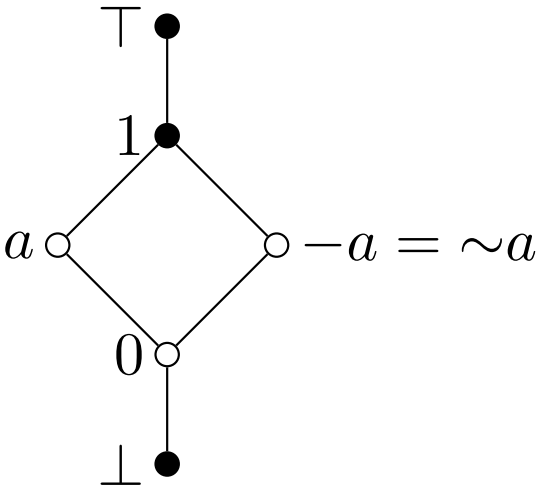
$$\sim a = a \backslash 0 \qquad \text{and} \quad -a = 0 / a.$$

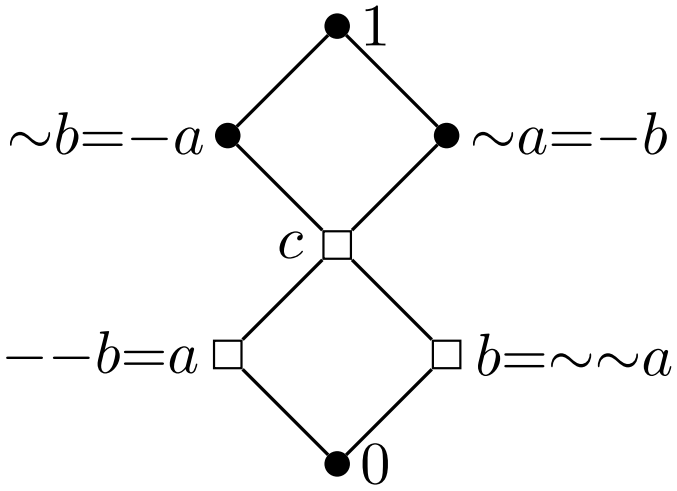
If $\sim - a = a = - \sim a$ for all $a \in A$, then $\langle A, \wedge, \vee, \cdot, 1, 0, \backslash, / \rangle$ is an InFL-algebra. A *distributive* InFL-algebra (DInFL-algebra) has a distributive lattice reduct.

NB: can axiomatize InFL-algebras with $\langle A, \wedge, \vee, \cdot, 1, \sim, - \rangle$.

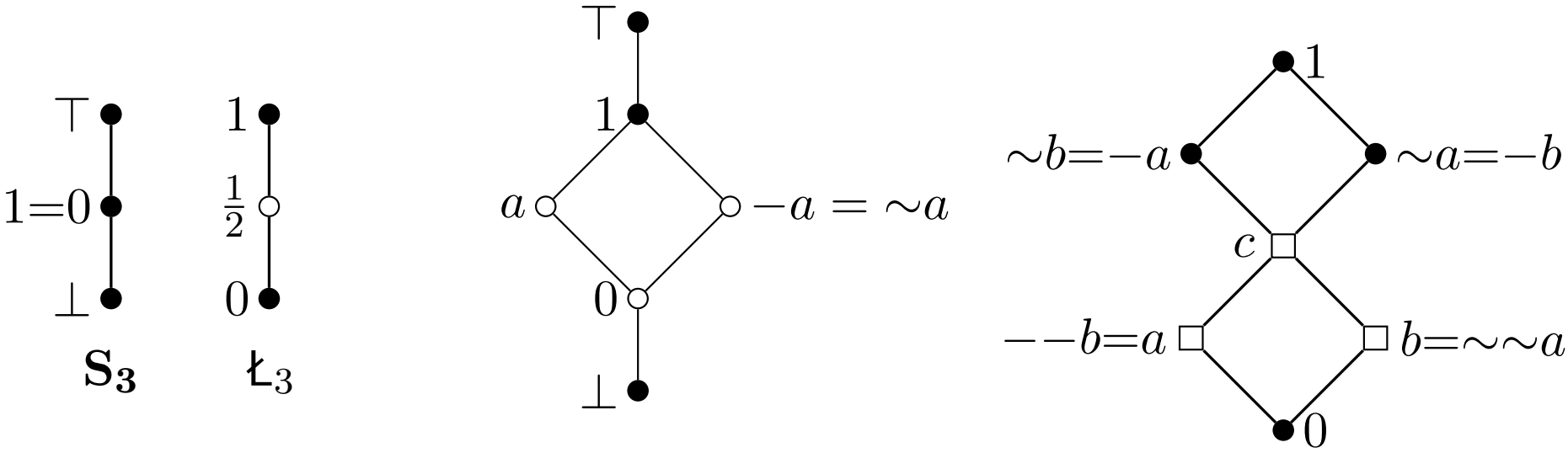
Examples of DInFL-algebras







Examples of DInFL-algebras



Example

For $\mathbf{A} = \langle A, \wedge, \vee, ', \perp, \top, \cdot, 1, ^{\sqcup} \rangle$ an RA, let $\sim a = -a = (a')^{\sqcup}$. Then $\langle A, \wedge, \vee, \cdot, 1, \sim, - \rangle$ is a DInFL-algebra.

Quasi relation algebras

Definition (Jipsen & Galatos 2013)

A quasi relation algebra (qRA) is an InFL-algebra with a unary operation $\neg$ satisfying:

$$\neg(a \vee b) = \neg a \wedge \neg b \qquad \neg\neg a = a$$

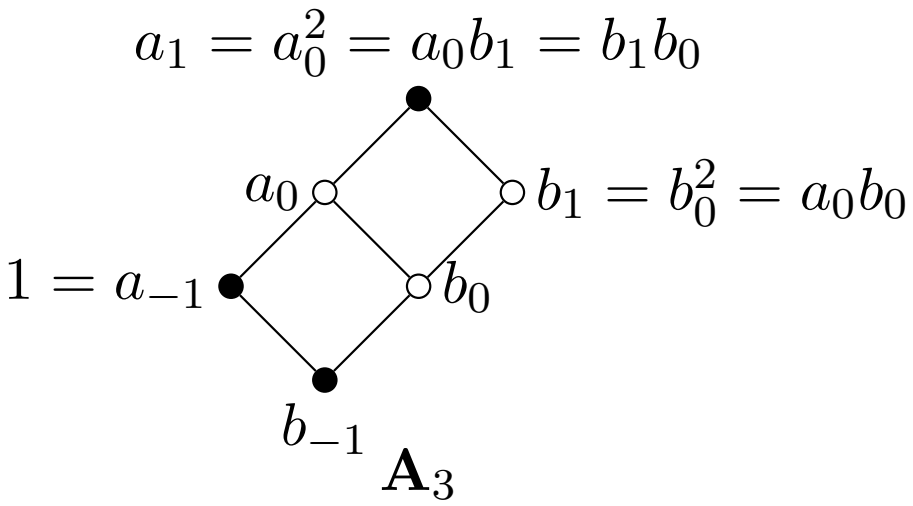
$$\neg(a \cdot b) = \sim(-\neg b \cdot -\neg a)$$

A qRA with a distributive lattice reduct will be called a DqRA.

Examples of DqRAs

Let $n \in \omega$ with $n \geqslant 3$. If $n = 2k+1$ for $k \geqslant 1$, set

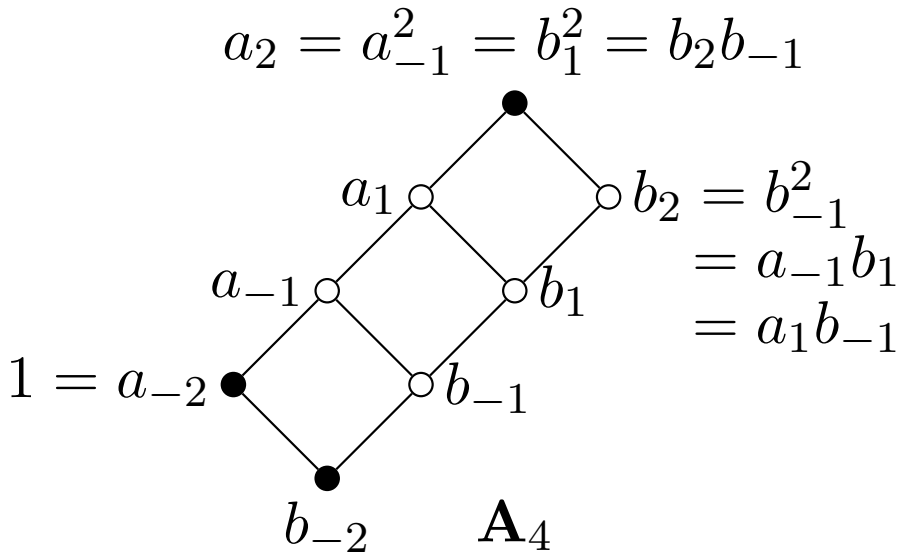
$$A_n = \{a_{-k}, \ldots, a_{-1}, a_0, a_1, \ldots, a_k, b_{-k}, \ldots, b_{-1}, b_0, b_1, \ldots, b_k\}.$$



Examples of DqRAs

Let $n \in \omega$ with $n \geqslant 3$. If $n = 2k$ for $k \geqslant 2$, set

$$A_n = \{a_{-k}, \ldots, a_{-1}, a_1, \ldots, a_k, b_{-k}, \ldots, b_{-1}, b_1, \ldots, b_k\},$$



Distributive residuated lattices from pomonoids

Definition

A partially ordered algebra $\mathbf{P} = \langle P, \leqslant, \cdot, 1 \rangle$ is a pomonoid if

- $(P, \leqslant)$ is a poset
- $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
- $a \cdot 1 = a = 1 \cdot a$
- if $a \leqslant b$ then $a \cdot c \leqslant b \cdot c$ and $c \cdot a \leqslant c \cdot b$

Distributive residuated lattices from pomonoids

Definition

A partially ordered algebra $\mathbf{P} = \langle P, \leqslant, \cdot, 1 \rangle$ is a pomonoid if

- $(P, \leqslant)$ is a poset
- $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
- $a \cdot 1 = a = 1 \cdot a$
- if $a \leqslant b$ then $a \cdot c \leqslant b \cdot c$ and $c \cdot a \leqslant c \cdot b$

For a pomonoid $\mathbf{P}$ and $U, V \in \mathsf{Up}(P, \leqslant)$ define

$$U \bullet V = \uparrow\{\, a \cdot b \mid a \in U, b \in V \,\}.$$

Distributive residuated lattices from pomonoids

Definition

A partially ordered algebra $\mathbf{P} = \langle P, \leqslant, \cdot, 1 \rangle$ is a pomonoid if

- $(P, \leqslant)$ is a poset
- $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
- $a \cdot 1 = a = 1 \cdot a$
- if $a \leqslant b$ then $a \cdot c \leqslant b \cdot c$ and $c \cdot a \leqslant c \cdot b$

For a pomonoid $\mathbf{P}$ and $U, V \in \mathsf{Up}(P, \leqslant)$ define

$$U \bullet V = \uparrow\{\, a \cdot b \mid a \in U, b \in V \,\}.$$

Theorem (Galatos 2003)

Let $\mathbf{P}$ be a pomonoid. Then

$$\mathcal{U}(\mathbf{P}) = \langle \mathsf{Up}(P, \leqslant), \cap, \cup, \bullet, \uparrow\{1\}, \backslash, / \rangle$$

is a distributive residuated lattice.

Representing DRLs with binary relations

Consider $\sigma : \mathcal{U}(\mathbf{P}) \to \langle \mathsf{Up}(P^{\partial} \times P), \cap, \cup, ;, \leqslant, \backslash, / \rangle$ defined by

$$\sigma(U) = \uparrow\{\, (g, g \cdot u) \mid g \in P, u \in U \,\}$$

Representing DRLs with binary relations

Consider $\sigma : \mathcal{U}(\mathbf{P}) \to \langle \mathsf{Up}(P^{\partial} \times P), \cap, \cup, ;, \leqslant, \backslash, / \rangle$ defined by

$$\sigma(U) = \uparrow\{\, (g, g \cdot u) \mid g \in P, u \in U \,\}$$

Proposition

For a pomonoid $\mathbf{P}$, the map σ preserves arbitrary joins, the monoid operation, and the monoid identity.

Representing DRLs with binary relations

Consider $\sigma : \mathcal{U}(\mathbf{P}) \to \langle \mathsf{Up}(P^\partial \times P), \cap, \cup, ;, \leqslant, \backslash, / \rangle$ defined by

$$\sigma(U) = \uparrow\{\, (g, g \cdot u) \mid g \in P, u \in U \,\}$$

Proposition

For a pomonoid $\mathbf{P}$, the map σ preserves arbitrary joins, the monoid operation, and the monoid identity.

Meets are *not* always preserved. Consider $\langle \{0,1\}, \leqslant, \cdot, 1 \rangle$ with $\leqslant = \{(0,0),(1,1)\}$ and $0 \cdot 0 = 0$.

Representing DRLs with binary relations

Consider $\sigma : \mathcal{U}(\mathbf{P}) \to \langle \mathsf{Up}(P^\partial \times P), \cap, \cup, ;, \leqslant, \backslash, / \rangle$ defined by

$$\sigma(U) = \uparrow\{\, (g, g \cdot u) \mid g \in P, u \in U \,\}$$

Proposition

For a pomonoid $\mathbf{P}$, the map σ preserves arbitrary joins, the monoid operation, and the monoid identity.

Meets are *not* always preserved. Consider $\langle \{0,1\}, \leqslant, \cdot, 1 \rangle$ with $\leqslant = \{(0,0),(1,1)\}$ and $0 \cdot 0 = 0$.

Proposition

Let $\langle P, \leqslant, \cdot, 1 \rangle$ be a pomonoid. The map σ preserves meets if and only if whenever $x \cdot u \leqslant y$ & $x \cdot v \leqslant y$, there exists w s.t. $u \leqslant w$, $v \leqslant w$ & $x \cdot w \leqslant y$.

ipo-monoids

Definition (GJKO 2007, GJL 2023)

An *involutive partially ordered monoid* (or *ipo-monoid*) is a structure $\mathbf{P} = \langle P, \leqslant, \cdot, 1, ^{-}, ^{\sim} \rangle$ s.t. $\langle P, \leqslant \rangle$ is a poset, $\langle P, \cdot, 1 \rangle$ is a monoid, and for all $x, y \in P$:

$$x \leqslant y \quad \text{iff} \quad x \cdot y^{\sim} \leqslant 1^{-} \quad \text{iff} \quad y^{-} \cdot x \leqslant 1^{-}.$$

DInFL-algebras from ipo-monoids

Let $\mathbf{P} = \langle P, \leqslant, \cdot, 1, ^{-}, ^{\sim} \rangle$ be an ipo-monoid and $U \in \mathsf{Up}\,(P, \leqslant)$.

DInFL-algebras from ipo-monoids

Let $\mathbf{P} = \langle P, \leqslant, \cdot, 1, ^{-}, ^{\sim} \rangle$ be an ipo-monoid and $U \in \mathsf{Up}\,(P, \leqslant)$.

Define $-U = \{x^{-} \mid x \notin U\}$ and $\sim U = \{x^{\sim} \mid x \notin U\}$.

DInFL-algebras from ipo-monoids

Let $\mathbf{P} = \langle P, \leqslant, \cdot, 1, ^{-}, ^{\sim} \rangle$ be an ipo-monoid and $U \in \mathsf{Up}\,(P, \leqslant)$.

Define $-U = \{x^{-} \mid x \notin U\}$ and $\sim U = \{x^{\sim} \mid x \notin U\}$.

Lemma

Let $\mathbf{P} = \langle P, \leqslant, \cdot, 1, ^{-}, ^{\sim} \rangle$ be an ipo-monoid. If $U \in \mathsf{Up}\,(P, \leqslant)$, then $\sim U \in \mathsf{Up}\,(P, \leqslant)$ and $-U \in \mathsf{Up}\,(P, \leqslant)$.

DInFL-algebras from ipo-monoids

Let $\mathbf{P} = \langle P, \leqslant, \cdot, 1, ^{-}, ^{\sim} \rangle$ be an ipo-monoid and $U \in \mathsf{Up}\,(P, \leqslant)$.

Define $-U = \{x^{-} \mid x \notin U\}$ and $\sim U = \{x^{\sim} \mid x \notin U\}$.

Lemma

Let $\mathbf{P} = \langle P, \leqslant, \cdot, 1, ^{-}, ^{\sim} \rangle$ be an ipo-monoid. If $U \in \mathsf{Up}\,(P, \leqslant)$, then $\sim U \in \mathsf{Up}\,(P, \leqslant)$ and $-U \in \mathsf{Up}\,(P, \leqslant)$.

Theorem

Let $\mathbf{P} = \langle P, \leqslant, \cdot, 1, ^{-}, ^{\sim} \rangle$ be an ipo-monoid. Then

$$\mathcal{D}(\mathbf{P}) = \langle \mathsf{Up}\,(P, \leqslant), \cap, \cup, \bullet, \uparrow 1, -, \sim \rangle$$

is a DInFL-algebra. Moreover, $\mathcal{D}(\mathbf{P})$ is cyclic iff $\mathbf{P}$ is cyclic.

Another nice property of ipo-monoids

Recall $\sigma : \mathcal{U}(\mathbf{P}) \to \langle \mathsf{Up}(P^\partial \times P), \cap, \cup, ;, \leqslant, \backslash, / \rangle$ defined by

$$\sigma(U) = \uparrow\{\, (g, g \cdot u) \mid g \in P, u \in U \,\}$$

Another nice property of ipo-monoids

Recall $\sigma : \mathcal{U}(\mathbf{P}) \to \langle \mathsf{Up}(P^\partial \times P), \cap, \cup, ;, \leqslant, \backslash, / \rangle$ defined by

$$\sigma(U) = \uparrow\{\, (g, g \cdot u) \mid g \in P, u \in U \,\}$$

Also recall:

Proposition

Let $\langle P, \leqslant, \cdot, 1 \rangle$ be a pomonoid. The map σ preserves meets if and only if whenever $x \cdot u \leqslant y$ & $x \cdot v \leqslant y$, there exists w s.t. $u \leqslant w$, $v \leqslant w$ & $x \cdot w \leqslant y$.

Another nice property of ipo-monoids

Recall $\sigma : \mathcal{U}(\mathbf{P}) \to \langle \mathsf{Up}(P^\partial \times P), \cap, \cup, ;, \leqslant, \backslash, / \rangle$ defined by

$$\sigma(U) = \uparrow\{\, (g, g \cdot u) \mid g \in P, u \in U \,\}$$

Also recall:

Proposition

Let $\langle P, \leqslant, \cdot, 1 \rangle$ be a pomonoid. The map σ preserves meets if and only if whenever $x \cdot u \leqslant y$ & $x \cdot v \leqslant y$, there exists w s.t. $u \leqslant w$, $v \leqslant w$ & $x \cdot w \leqslant y$.

If $\mathbf{P} = \langle P, \leqslant, \cdot, 1, ^{-}, ^{\sim} \rangle$ is an ipo-monoid, then it satisfies the condition above.

Proof: $w = x \backslash y$ (since ipo-monoids are residuated).

Representable DInFL-algebras

$(x, y) \preccurlyeq (u, v)$ iff $u \leqslant x$ and $y \leqslant v$

Representable DInFL-algebras

$(x, y) \preccurlyeq (u, v)$ iff $u \leqslant x$ and $y \leqslant v$

Theorem (C., Robinson 2025)

Let $\mathbf{X} = \langle X, \leqslant \rangle$ be a poset and $\alpha : X \to X$ an order automorphism of $\mathbf{X}$. Set $1 = \leqslant$ and for $R \in \mathsf{Up}(X^2, \preccurlyeq)$, define $\sim R = R^{c\smile} ; \alpha$ and $-R = \alpha ; R^{c\smile}$. Then

$$\langle \mathsf{Up}(X^2, \preccurlyeq), \cap, \cup, ;, 1, -, \sim \rangle$$

is a DInFL-algebra and it is cyclic iff α is the identity.

Representable DInFL-algebras

$(x, y) \preccurlyeq (u, v)$ iff $u \leqslant x$ and $y \leqslant v$

Theorem (C., Robinson 2025)

Let $\mathbf{X} = \langle X, \leqslant \rangle$ be a poset and $\alpha : X \to X$ an order automorphism of $\mathbf{X}$. Set $1 = \leqslant$ and for $R \in \mathsf{Up}(X^2, \preccurlyeq)$, define $\sim R = R^{c\smile} ; \alpha$ and $-R = \alpha ; R^{c\smile}$. Then

$$\langle \mathsf{Up}(X^2, \preccurlyeq), \cap, \cup, ;, 1, -, \sim \rangle$$

is a DInFL-algebra and it is cyclic iff α is the identity.

Algebras of the form described above are full DInFL-algebras. Class denoted by FDInFL.

Representable DInFL-algebras

$(x, y) \preccurlyeq (u, v)$ iff $u \leqslant x$ and $y \leqslant v$

Theorem (C., Robinson 2025)

Let $\mathbf{X} = \langle X, \leqslant \rangle$ be a poset and $\alpha : X \to X$ an order automorphism of $\mathbf{X}$. Set $1 = \leqslant$ and for $R \in \mathsf{Up}(X^2, \preccurlyeq)$, define $\sim R = R^{c\smile} ; \alpha$ and $-R = \alpha ; R^{c\smile}$. Then

$$\langle \mathsf{Up}(X^2, \preccurlyeq), \cap, \cup, ;, 1, -, \sim \rangle$$

is a DInFL-algebra and it is cyclic iff α is the identity.

Algebras of the form described above are full DInFL-algebras. Class denoted by FDInFL.

Definition (C., Robinson 2025)

A DInFL-algebra $\mathbf{A} = \langle A, \wedge, \vee, \cdot, 1, -, \sim, \rangle$ will be called representable if $\mathbf{A} \in \mathbb{ISP}(\mathsf{FDInFL})$.

DInFL-algebras of binary relations

Theorem

Let $\mathbf{P} = \langle P, \leqslant, \cdot, 1, ^{-}, ^{\sim} \rangle$ *be an ipo-monoid. Then the algebra of binary relations* $\left\langle \mathsf{Up}\left(P^2, \preccurlyeq\right), \cap, \cup, ;, \leqslant, -, \sim \right\rangle$ *is a DInFL-algebra.*

Proof.

Define $\alpha : P \to P$ by $\alpha(x) = x^{\sim\sim}$. Then α is an order isomorphism. $\square$

DInFL-algebras of binary relations

Theorem

Let $\mathbf{P} = \langle P, \leqslant, \cdot, 1, ^{-}, ^{\sim} \rangle$ *be an ipo-monoid. Then the algebra of binary relations* $\left\langle \mathsf{Up}\left(P^2, \preccurlyeq\right), \cap, \cup, ;, \leqslant, -, \sim \right\rangle$ *is a DInFL-algebra.*

Proof.

Define $\alpha : P \to P$ by $\alpha(x) = x^{\sim\sim}$. Then α is an order isomorphism. $\square$

NB: we can't yet show that $\sigma : \mathsf{Up}(P, \leqslant) \to \mathsf{Up}(P^2, \preccurlyeq)$ is a DInFL-algebra embedding.

Pregroups

Definition (Lambek 1999)

A partially ordered algebra $\langle P, \leqslant, \cdot, 1, ^{\ell}, ^{r} \rangle$ is a pregroup if

- $\langle P, \leqslant, \cdot, 1 \rangle$ is a pomonoid
- $x^{\ell} \cdot x \leqslant 1 \leqslant x \cdot x^{\ell}$ and $x \cdot x^{r} \leqslant 1 \leqslant x^{r} \cdot x$

Pregroups

Definition (Lambek 1999)

A partially ordered algebra $\langle P, \leqslant, \cdot, 1, ^{\ell}, ^{r} \rangle$ is a pregroup if

- $\langle P, \leqslant, \cdot, 1 \rangle$ is a pomonoid
- $x^{\ell} \cdot x \leqslant 1 \leqslant x \cdot x^{\ell}$ and $x \cdot x^{r} \leqslant 1 \leqslant x^{r} \cdot x$

Example

Any group $\mathbf{G} = \langle G, \cdot, ^{-1}, e \rangle$ is a pregroup with $\leqslant = \mathrm{id}_G$ and $a^r = a^{\ell} = a^{-1}$.

Pregroups

Definition (Lambek 1999)

A partially ordered algebra $\langle P, \leqslant, \cdot, 1, ^{\ell}, ^{r} \rangle$ is a pregroup if

- $\langle P, \leqslant, \cdot, 1 \rangle$ is a pomonoid
- $x^{\ell} \cdot x \leqslant 1 \leqslant x \cdot x^{\ell}$ and $x \cdot x^{r} \leqslant 1 \leqslant x^{r} \cdot x$

Example

Any group $\mathbf{G} = \langle G, \cdot, ^{-1}, e \rangle$ is a pregroup with $\leqslant = \mathrm{id}_G$ and $a^r = a^\ell = a^{-1}$.

Fact: all *finite* pregroups come from finite groups (and hence have $a^r = a^\ell$ and the discrete order).

Pregroups

Definition (Lambek 1999)

A partially ordered algebra $\langle P, \leqslant, \cdot, 1, ^{\ell}, ^{r} \rangle$ is a pregroup if

- $\langle P, \leqslant, \cdot, 1 \rangle$ is a pomonoid
- $x^{\ell} \cdot x \leqslant 1 \leqslant x \cdot x^{\ell}$ and $x \cdot x^{r} \leqslant 1 \leqslant x^{r} \cdot x$

Example

Any group $\mathbf{G} = \langle G, \cdot, ^{-1}, e \rangle$ is a pregroup with $\leqslant = \mathrm{id}_G$ and $a^r = a^\ell = a^{-1}$.

Fact: all *finite* pregroups come from finite groups (and hence have $a^r = a^\ell$ and the discrete order).

Example (Lambek pregroup)

Consider the set of all unbounded, monotone functions $f : \mathbb{Z} \to \mathbb{Z}$.

$$f^{\ell}(x) = \min\{y \in \mathbb{Z} \mid x \leqslant f(y)\},\ f^{r}(x) = \max\{y \in \mathbb{Z} \mid f(y) \leqslant x\}$$

Representable DInFL-algebras from pregroups

We have the following facts and their consquences:

- Every pregroup $\mathbf{P}$ is an ipo-monoid.

Representable DInFL-algebras from pregroups

We have the following facts and their consquences:

- Every pregroup $\mathbf{P}$ is an ipo-monoid.
- Hence $\mathcal{D}(\mathbf{P})$ is a DInFL-algebra and $\left\langle \mathsf{Up}\left(P^2, \preccurlyeq\right), \cap, \cup, ;, \leqslant, -, \sim \right\rangle$ is a DInFL-algebra. Here $\alpha(x) = x^{rr}$.
- σ preserves meets.

Representable DInFL-algebras from pregroups

We have the following facts and their consquences:

- Every pregroup $\mathbf{P}$ is an ipo-monoid.
- Hence $\mathcal{D}(\mathbf{P})$ is a DInFL-algebra and $\left\langle \mathsf{Up}\left(P^2, \preccurlyeq\right), \cap, \cup, ;, \leqslant, -, \sim \right\rangle$ is a DInFL-algebra. Here $\alpha(x) = x^{rr}$.
- σ preserves meets.
- σ preserves $\sim$ and $-$.

Representable DInFL-algebras from pregroups

We have the following facts and their consquences:

- Every pregroup $\mathbf{P}$ is an ipo-monoid.
- Hence $\mathcal{D}(\mathbf{P})$ is a DInFL-algebra and $\left\langle \mathsf{Up}\left(P^2, \preccurlyeq\right), \cap, \cup, ;, \leqslant, -, \sim \right\rangle$ is a DInFL-algebra. Here $\alpha(x) = x^{rr}$.
- σ preserves meets.
- σ preserves $\sim$ and $-$.

Moreover, if $\mathbf{P}$ is an ipo-monoid and σ preserves $\sim$ and $-$, then $\mathbf{P}$ is a pregroup.

Representable DInFL-algebras from pregroups

We have the following facts and their consquences:

- Every pregroup $\mathbf{P}$ is an ipo-monoid.
- Hence $\mathcal{D}(\mathbf{P})$ is a DInFL-algebra and $\left\langle \mathrm{Up}\left(P^2, \preccurlyeq\right), \cap, \cup, ;, \leqslant, -, \sim \right\rangle$ is a DInFL-algebra. Here $\alpha(x) = x^{rr}$.
- σ preserves meets.
- σ preserves $\sim$ and $-$.

Moreover, if $\mathbf{P}$ is an ipo-monoid and σ preserves $\sim$ and $-$, then $\mathbf{P}$ is a pregroup.

Theorem

Let $\mathbf{A}$ be a DInFL-algebra. If there exists a pregroup $\mathbf{P}$ such that $\mathbf{A} \approx \mathcal{D}(\mathbf{P})$, then $\mathbf{A}$ is representable.

Theorem

Let $\mathbf{X} = \langle X, \leqslant \rangle$ *be a poset and* $\alpha : X \to X$ *an order automorphism of* $\mathbf{X}$*. Set* $1 = \leqslant$ *and for* $R \in \mathsf{Up}(X^2, \preccurlyeq)$*, define* $\sim R = R^{c\smile} ; \alpha$ *and* $-R = \alpha ; R^{c\smile}$*. Further, if* $\beta : X \to X$ *is a self-inverse dual order automorphism of* $\mathbf{X}$ *such that* $\beta = \alpha ; \beta ; \alpha$*, then defining* $\neg R = \alpha ; \beta ; R^c ; \beta$ *we get that*

$$\langle \mathsf{Up}(X^2, \preccurlyeq), \cap, \cup, ;, 1, -, \sim, \neg \rangle$$

is a distributive quasi relation algebra.

Algebras of the form described above are full DqRAs. Classes denoted by FDqRA.

Definition (C., Robinson 2025)

A DqRA $\mathbf{B} = \langle B, \wedge, \vee, \cdot, 1, -, \sim, \neg \rangle$ is *representable* if $\mathbf{B} \in \mathbb{ISP}(\mathsf{FDqRA})$.

Ortho pregroups and DqRAs

Definition (Jipsen & Galatos 2013)

A quasi relation algebra (qRA) is an InFL-algebra with a unary operation $\neg$ satisfying:

$$\neg(a \vee b) = \neg a \wedge \neg b \qquad \neg\neg a = a$$

$$\neg(a \cdot b) = \sim(-\neg b \cdot -\neg a)$$

Ortho pregroups and DqRAs

Definition (Jipsen & Galatos 2013)

A quasi relation algebra (qRA) is an InFL-algebra with a unary operation $\neg$ satisfying:

$$\neg(a \vee b) = \neg a \wedge \neg b \qquad \neg\neg a = a$$

$$\neg(a \cdot b) = \sim(-\neg b \cdot -\neg a)$$

We want relational representations for distributive quasi relation algebras (DqRAs).

Ortho pregroups and DqRAs

Definition (Jipsen & Galatos 2013)

A quasi relation algebra (qRA) is an InFL-algebra with a unary operation $\neg$ satisfying:

$$\neg(a \vee b) = \neg a \wedge \neg b \qquad \neg\neg a = a$$

$$\neg(a \cdot b) = \sim(-\neg b \cdot -\neg a)$$

We want relational representations for distributive quasi relation algebras (DqRAs).

Definition

$\mathbf{P} = \langle P, \leqslant, \cdot, 1, {}^{\ell}, {}^{r}, {}^{\neg} \rangle$ is an ortho pregroup if $x^{\neg\neg} = x$ and $xy \leqslant z^{\ell}$ iff $y^{r\neg}x^{r\neg} \leqslant z^{\neg}$.

Ortho pregroups and DqRAs

Definition (Jipsen & Galatos 2013)

A quasi relation algebra (qRA) is an InFL-algebra with a unary operation $\neg$ satisfying:

$$\neg(a \vee b) = \neg a \wedge \neg b \qquad \neg\neg a = a$$

$$\neg(a \cdot b) = \sim(-\neg b \cdot -\neg a)$$

We want relational representations for distributive quasi relation algebras (DqRAs).

Definition

$\mathbf{P} = \langle P, \leqslant, \cdot, 1, {}^{\ell}, {}^{r}, {}^{\neg} \rangle$ is an ortho pregroup if $x^{\neg\neg} = x$ and $xy \leqslant z^{\ell}$ iff $y^{r\neg}x^{r\neg} \leqslant z^{\neg}$.

Example

If $\mathbf{G} = \langle G, \cdot, {}^{-1}, e \rangle$ is a group and $\mathrm{id} : G \to G$ the identity function, then $\langle G, =, \cdot, e, {}^{-1}, {}^{-1}, \mathrm{id} \rangle$ is an ortho pregroup.

More examples of ortho pregroups

Definition

$\mathbf{P} = \langle P, \leqslant, \cdot, 1, {}^{\ell}, {}^{r}, {}^{\neg} \rangle$ is an ortho pregroup if $x^{\neg\neg} = x$ and $xy \leqslant z^{\ell}$ iff $y^{r\neg}x^{r\neg} \leqslant z^{\neg}$.

More examples of ortho pregroups

Definition

$\mathbf{P} = \langle P, \leqslant, \cdot, 1, ^{\ell}, ^{r}, ^{\neg} \rangle$ is an ortho pregroup if $x^{\neg\neg} = x$ and $xy \leqslant z^{\ell}$ iff $y^{r\neg}x^{r\neg} \leqslant z^{\neg}$.

Example

Let $\mathbf{G} = \langle G, \cdot, ^{-1}, e \rangle$ be an Abelian group. Then $\langle G, =, \cdot, e, ^{-1}, ^{-1}, ^{-1} \rangle$ is an ortho pregroup.

More examples of ortho pregroups

Definition

$\mathbf{P} = \langle P, \leqslant, \cdot, 1, ^{\ell}, ^{r}, ^{\neg} \rangle$ is an ortho pregroup if $x^{\neg\neg} = x$ and $xy \leqslant z^{\ell}$ iff $y^{r\neg}x^{r\neg} \leqslant z^{\neg}$.

Example

Let $\mathbf{G} = \langle G, \cdot, ^{-1}, e \rangle$ be an Abelian group. Then $\langle G, =, \cdot, e, ^{-1}, ^{-1}, ^{-1} \rangle$ is an ortho pregroup.

Recall the Lambek pregroup:

Example (Lambek pregroup)

Consider the set of all unbounded, monotone functions $f : \mathbb{Z} \to \mathbb{Z}$.

$$f^{\ell}(x) = \min\{y \in \mathbb{Z} \mid x \leqslant f(y)\},\ f^{r}(x) = \max\{y \in \mathbb{Z} \mid f(y) \leqslant x\}$$

Now let $f^{\neg}(x) = -f(-x)$. This gives us an ortho pregroup.

Representing DqRAs with binary relations

Theorem

Let $\mathbf{P} = \langle P, \leqslant, \cdot, 1, {}^{\ell}, {}^{r}, {}^{\neg} \rangle$ *be an ortho pregroup. Define* $\neg U = \{\, a^{\neg} \mid a \notin U \,\}$ *for* $U \in \mathsf{Up}(P, \leqslant)$*. Then*

$$\mathcal{Q}(\mathbf{P}) = \langle \mathsf{Up}(P, \leqslant), \cap, \cup, \bullet, \uparrow\{1\}, \sim, -, \neg \rangle$$

is a DqRA.

Representing DqRAs with binary relations

Theorem

Let $\mathbf{P} = \langle P, \leqslant, \cdot, 1, {}^{\ell}, {}^{r}, {}^{\neg} \rangle$ *be an ortho pregroup. Define* $\neg U = \{\, a^{\neg} \mid a \notin U \,\}$ *for* $U \in \mathsf{Up}(P, \leqslant)$*. Then*

$$\mathcal{Q}(\mathbf{P}) = \langle \mathsf{Up}(P, \leqslant), \cap, \cup, \bullet, \uparrow\{1\}, \sim, -, \neg \rangle$$

is a DqRA.

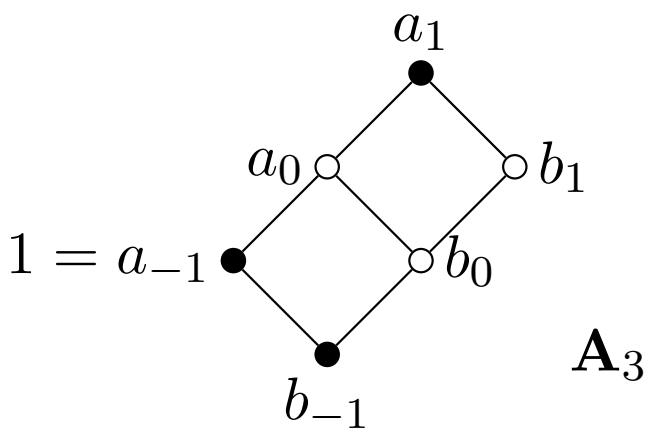
Theorem

Let $\mathbf{A}$ *be a DqRA. If there exists an involutive pregroup* $\mathbf{P}$ *with* $\mathbf{A} \approx \mathcal{Q}(\mathbf{P})$*, then* $\mathbf{A}$ *is representable.*

Proof.

Define $\beta : P \to P$ by $\beta(x) = x^{\neg}$ and use the embedding σ. $\square$

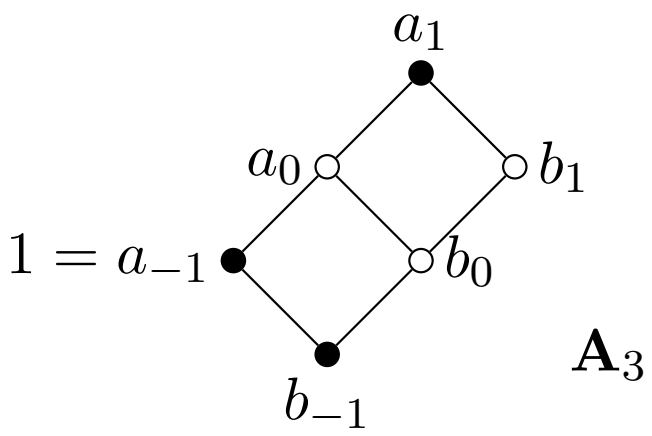
Examples from products of groups



Theorem

Let $n \geqslant 3$. Then the DqRA $\mathbf{A}_n$ is representable.

Examples from products of groups



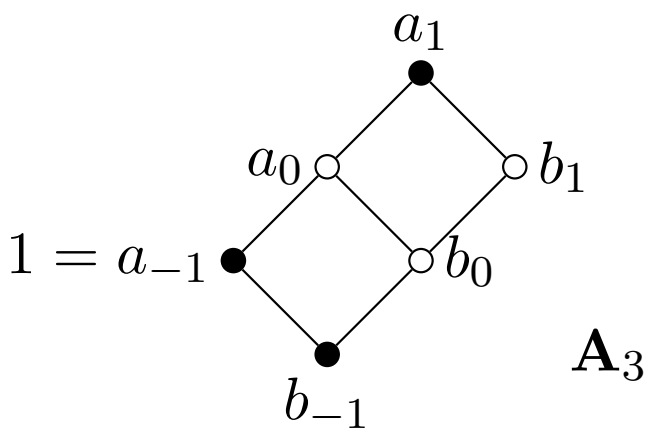
Theorem

Let $n \geqslant 3$. Then the DqRA $\mathbf{A}_n$ is representable.

Proof.

For $n = 3$, consider the orthopregroup $\mathbf{Z}_7 = \langle \mathbb{Z}_7, =, +, 0, -, -, - \rangle$. We get $\psi : \mathbf{A}_3 \hookrightarrow \mathcal{Q}(\mathbf{Z}_7)$.

Examples from products of groups



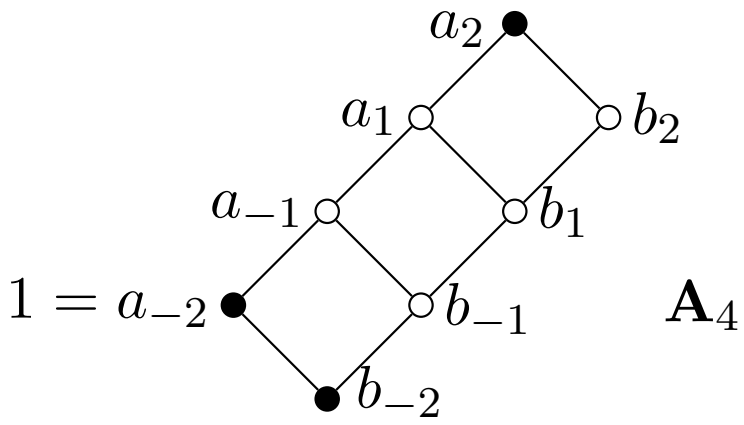
Theorem

Let $n \geqslant 3$. Then the DqRA $\mathbf{A}_n$ is representable.

Proof.

For $n = 3$, consider the orthopregroup $\mathbf{Z}_7 = \langle \mathbb{Z}_7, =, +, 0, -, -, - \rangle$.
We get $\psi : \mathbf{A}_3 \hookrightarrow \mathcal{Q}(\mathbf{Z}_7)$.
$\psi(b_{-1}) = \emptyset$, $\psi(a_{-1}) = \{0\}$, $\psi(b_0) = \{1, 2, 4\}$, $\psi(b_1) = \mathbb{Z}_7 \backslash \{0\}$ $\square$

Examples from products of groups cont...



Proof.

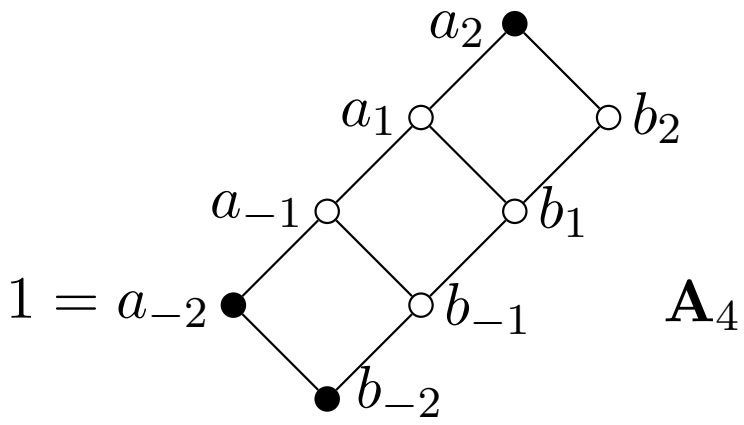
Sketch: For $n \geqslant 4$, consider the ortho pregroup

$$\mathbf{Z}_{n-2} \times \mathbf{Z}_7 = \langle \mathbb{Z}_{n-2} \times \mathbb{Z}_7, =, +, (0,0), -, -, - \rangle.$$

$$T = \{(0,1), (0,2), (0,4)\} \cup \{(m, \ell) \mid 1 \leqslant m \leqslant n-3, \ell \in \{3,5,6\}\}.$$

Use T to define the images of the b_i under an embedding $\psi : A_n \to \mathsf{Up}(\mathbb{Z}_{n-2} \times \mathbb{Z}_7, =)$.

Examples from products of groups cont...



Proof.

Sketch: For $n \geqslant 4$, consider the ortho pregroup

$$\mathbf{Z}_{n-2} \times \mathbf{Z}_7 = \langle \mathbb{Z}_{n-2} \times \mathbb{Z}_7, =, +, (0,0), -, -, - \rangle.$$

$$T = \{(0,1), (0,2), (0,4)\} \cup \{(m, \ell) \mid 1 \leqslant m \leqslant n-3, \ell \in \{3,5,6\}\}.$$

Use T to define the images of the b_i under an embedding $\psi : A_n \to \mathsf{Up}(\mathbb{Z}_{n-2} \times \mathbb{Z}_7, =)$. Then, let $\psi(a_i) = \psi(b_i) \cup \{(0,0)\}$. $\square$

Ongoing work

- Describe possible $^\neg$ operations on those pregroups which come from groups. This would lead to a description of all possible finite involutive pregroups.
- The condition required for σ to preserve meets is also satisfied by lattice-ordered monoids. What do we know about the residuated lattice $\langle \mathsf{Up}(P, \leqslant), \cap, \cup, \bullet, \uparrow\{1\}, \backslash, / \rangle$ when $\mathbf{P}$ is an ℓ-monoid?
- Extend to (first define!) pregroupoids to emulate Brandt groupoids of Jónsson and Tarski. See also Jipsen (2017).
- Adapt the methods of Andréka, Nemeti & Givant (2020) to find *non-representable* DInFL-algebras & DqRAs.

References (1/2)

1. H. Andréka, I. Nemeti, S. Givant: Nonrepresentable relation algebras from groups, *Rev. Symb. Log.* **13**, 861–881 (2020)
2. L.H. Chin, A. Tarski: Distributive and modular laws in the arithmetic of relation algebras, *Univ. California Publ. Math.* **1**, 341–384 (1951)
3. A. Craig, C. Robinson: Representable distributive quasi relation algebras, *Algebra Universalis*, 86:12 (2025)
4. N. Galatos: Varieties of residuated lattices, PhD thesis, Vanderbilt University (2003)
5. N. Galatos, P. Jipsen: Relation algebras as expanded FL-algebras, *Algebra Universalis* **69**, 1–21 (2013)
6. N. Galatos, P. Jipsen, T. Kowalski, H. Ono, Residuated Lattices: An Algebraic Glimpse at Substructural Logics, Elsevier (2007)

References (2/2)

7. Gil-Férez, J., Jipsen, P., Lodhia, S.: The structure of locally integral involutive po-monoids and semirings, In: R. Glück et al. (Eds.): RAMiCS 2023, LNCS 13896, pp. 69–86, 2023.
8. Jipsen, P.: Relation algebras, idempotent semirings and generalized bunched implication algebras, In: RAMiCS 2017, P. Höfner et al. (Eds.), LNCS 10226, 144–158 (2017).
9. B. Jónsson and A. Tarski: Representation problems for relation algebras, *Bull. AMS* **54**, 89 (1948)
10. B. Jónsson and A. Tarski: Boolean algebras with operators `II`, *Amer. J. Math.* **74**, 127–162 (1952)
11. J. Lambek: Type grammars revisited, In: *Logical Aspects of Computational Linguistics*, Lecomte et al. (Eds), LNAI 1582, 1–27 (1999)